Jie Wu

PhD student in Physics

College of Physics, Chongqing University, Chongqing 401331, China

ORCID | Web of Science | wujie3375.github.io | wujie3375@gmail.com
INSPIRE | ResearchGate | Google Scholar | Astrophysics Data System

About me

I am a Ph.D. student in Physics at Chongqing University. My research focuses on gravitational-wave data analysis, waveform modeling, and detector performance for space- and ground-based detectors. I am particularly interested in signal simulation, parameter estimation, and tests of gravitation with compact-binary systems, with broader connections to astrophysics and cosmology.

Research

Gravitational waves	Waveform modeling, signal simulation.
Data analysis	Parameter estimation, Fisher matrix, Bayesian analysis.
Detectors	Space- and ground-based detectors, time-delay interferometry, multi-messenger observation.
Astrophysical sources	Binary black holes, galactic binaries, hierarchical triple systems.
Gravitation	Modified gravity, cosmology, dark matter, dynamical friction.

Education

Chongqing University — *Chongqing, China*

Ph.D. in Physics — *Sep 2022 - Jun 2028 (expected)*

- Advisor: Prof. Jin Li
- Research: Gravitational wave data simulation and detector performance evaluation

China West Normal University — *Nanchong, China*

B.S. in Physics — *Sep 2018 - Jun 2022*

- Advisors: Assoc. Prof. Di Wu and Assoc. Prof. Guo-Ping Li
- Research: Ground-based gravitational wave detection and data processing
- Thesis (in Chinese): An Analysis of the LIGO Gravitational Waves Data Based on Newtonian Approximate Model

Experience

Beijing Normal University — *Beijing, China*

Visitor — *Feb 2024 - Apr 2024*

- Advisor: Prof. Zhoujian Cao
- Research: Gravitational wave waveform simulation and moving source effect

University of Chinese Academy of Sciences — *Beijing, China*

Participant — *Sep 2020 - Sep 2021*

- Advisor: Assoc. Prof. Yong Tang
- Research: Analysis of gravitational wave data
- Program: College Student Innovation and Practice Program

Skills

Languages	Chinese (native), English.
Programming	Python, Mathematica, MATLAB
Data Analysis	Experienced in handling and analyzing datasets (statistical analysis, data visualization, and signal processing).
Teaching	High School Physics Teacher Qualification Certificate.

Honors & Awards

Awards

- 2025.12 **Doctoral Student Program of the Young S&T Talents Cultivation Project**, CAST
- 2025.12 **Advanced Individual in Scientific and Academic Innovation**, Chongqing U.
- 2023.12 **Second Prize (Ranked 2nd-3rd)**, The 7th Sichuan Chongqing Astronomy Competition
- 2022.06 **Excellent Graduation Thesis**, China West Normal U.
- 2022.05 **Outstanding Graduate**, China West Normal U.
- 2018.11 **Third Prize (Ranked 7th-8th)**, The 5th Sichuan Chongqing Astronomy Competition

Scholarships

- 2025.12 **National Scholarship**, Ministry of Education of China
- 2024.09 **Theoretical Physics Graduate Scholarship (Twice)**, Chongqing U.
- 2022-2023 **Graduate Academic Scholarship (Twice)**, Chongqing U.
- 2020-2022 **First-class Scholarship (Three times)**, China West Normal U.
- 2020.12 **Haotian Astronomy Scholarship**, Nanjing VasTech Astronomical Instrument & Equipment Co. Ltd.
- 2018-2021 **Second-class Scholarship (Four times)**, China West Normal U.

Publications

My publication record currently includes 16 papers, 10 of which are first-author papers.
Publications are listed in reverse chronological order (*: corresponding author; †: these authors contributed equally).

First-author

[1] **J. Wu**, M. Sun, J. Li* and Z. Cao*, Constraining AGN Disk Properties with Gravitational Waves from Inspiraling Stellar-Mass Binary Black Holes in Hierarchical Triple Systems, *Sci. China Phys. Mech. Astron.* accepted (2026), arXiv:2609.00945.

[2] **J. Wu**, M. Sun and J. Li*, Tests of general relativity using analytic derivatives of parametrized post-Einsteinian gravitational waveforms within the Fisher-matrix framework, *Eur. Phys. J. C* accepted (2026), arXiv:2607.04238.

[3] **J. Wu**, J.-T. Yao, M. Sun, J. Li* and Z. Cao*, Constraining Kerr supermassive black hole properties using gravitational waves from inspiraling stellar-mass binary black holes, *Phys. Rev. D* **113**, 124024 (2026), arXiv:2606.12784.

[4] **J. Wu**, M. Sun and J. Li*, Constraints and detection capabilities of GW polarizations with space-based detectors in different TDI combinations, *Class. Quant. Grav.* **43**, 065017 (2026), arXiv:2411.03631.

[5] **J. Wu**, Y. Xiao, M. Sun and J. Li*, Probing globular clusters using modulated gravitational waves from binary black holes, (2025), arXiv:2508.04021 (under review).

[6] **J. Wu**, M. Sun, X. Ma, X. Liu, J. Li* and Z. Cao*, Effect of kick velocity on gravitational wave detection of binary black holes with space- and ground-based detectors, *Phys. Rev. D* **112**, 024040 (2025), arXiv:2502.13710.

[7] **J. Wu** and J. Li*, Prospects of constraining on the polarizations of gravitational waves from binary black holes using space- and ground-based detectors, *Phys. Rev. D* **110**, 084057 (2024), arXiv:2407.13590.

[8] **J. Wu**, J. Li*, X. Liu and Z. Cao, Comparison and application of different post-Newtonian models for inspiralling stellar-mass binary black holes with space-based GW detectors, *Phys. Rev. D* **109**, 104014 (2024), arXiv:2401.03113.

[9] **J. Wu** and J. Li*, Subtraction of the confusion foreground and parameter uncertainty of resolvable galactic binaries on the networks of space-based gravitational-wave detectors, *Phys. Rev. D* **108**, 124047 (2023), arXiv:2307.05568.

[10] **J. Wu**, J. Li* and Q.-Q. Jiang*, Application of Newtonian approximate model to LIGO gravitational wave data processing, *Chin. Phys. B* **32**, 090401 (2023).

Co-author

[11] M. Sun†, B. Wang†, **J. Wu**, J. Li* and S. Ye, Effects of Solar Wind Plasma Noise on Stochastic Gravitational Wave Background Searches with the LISA-Taiji Network, (2026), arXiv:2607.05368 (under review).

[12] M. Sun, **J. Wu**, Q. Hu, J. Li*, N. Yang, X. Ma, B. Wang, M. Zhang and Y. Zhong*, Identifiability of g mode Resonances in Eccentric Binary Neutron Stars with Multidetector Observations, (2026), arXiv:2606.11959 (under review).

[13] M. Sun, **J. Wu**, Q. Hu, J. Li*, N. Yang, X. Ma, B. Wang, M. Zhang and Y. Zhong*, Detection of multiband lensed gravitational waves from dark matter halos with deep learning, *Phys. Rev. D* **113**, 103004 (2026), arXiv:2511.09107.

[14] M. Sun, **J. Wu**, J. Li*, B. McCane, N. Yang, X. Ma, B. Wang and M. Zhang, Conditional Autoencoder for Generating Binary Neutron Star Waveforms with Tidal and Precession Effects, *Phys. Rev. D* **112**, 084016 (2025), arXiv:2503.19512.

[15] X. Ma, B. Wang, N. Yang, J. Li*, B. McCane, M. Sun, **J. Wu**, M. Zhang and Y. Meng*, Identification of Stochastic Gravitational Wave Backgrounds from Cosmic String Using Machine Learning, *Phys. Rev. D* **112**, 064081 (2025), arXiv:2502.11804.

[16] Y. Hu, **J. Wu**, H. Luo, G. Su, X. Meng, L. Liu and G. Chen*, Parallel manipulation of multiple ink droplets via near-infrared light on lubricant infused surface, *Appl. Phys. Lett.* **126**, 021602 (2025).